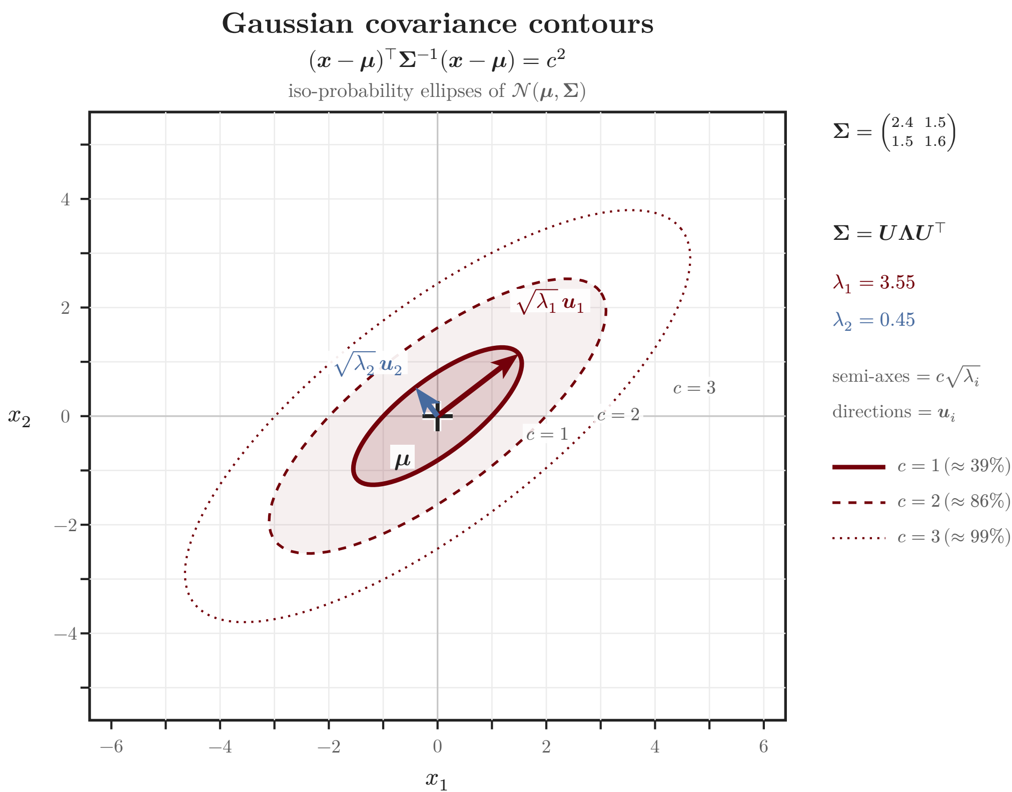

Gaussian covariance contours
$(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}) = c^2$
iso-probability ellipses of $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
$\boldsymbol{\Sigma} = \begin{pmatrix} 2.4 & 1.5 \\ 1.5 & 1.6 \end{pmatrix}$
$\boldsymbol{\Sigma} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$
$\lambda_1 = 3.55$
$\lambda_2 = 0.45$
semi-axes $= c\sqrt{\lambda_i}$
directions $= \boldsymbol{u}_i$
$c = 1\,(\approx 39\%)$
$c = 2\,(\approx 86\%)$
$c = 3\,(\approx 99\%)$
$\sqrt{\lambda_1}\,\boldsymbol{u}_1$
$\sqrt{\lambda_2}\,\boldsymbol{u}_2$
$\boldsymbol{\mu}$
$c = 1$
$c = 2$
$c = 3$
x_1
x_2
−6
−4
−2
0
2
4
6